

Week 1: Design Pattern and Principles

Exercise 1: Implementing the Singleton Pattern

```
//.Logger class

public class Logger {

    private static Logger instance;

    private Logger() {

    }

    public static Logger getInstance() {

        if (instance == null) {

            instance = new Logger();

        }

        return instance;

    }

    public void log(String message) {

        System.out.println("Log: " + message);

    }

}

//. SingletonPatternTest class

public class SingletonPatternTest{

    public static void main(String[] args) {

        Logger logger1 = Logger.getInstance();

        Logger logger2 = Logger.getInstance();

        logger1.log("This is a log message from logger1.");

        logger2.log("This is a log message from logger2.");

        System.out.println(logger1.hashCode());

        System.out.println(logger2.hashCode());

        System.out.println("Are both loggers the same instance? " + (logger1 == logger2));

    }

}
```

```
}  
}
```

Output:

```
PS C:\Users\SOWJANYA> & 'C:\Pro_ Open folder in new window (ctrl + click) e' '-agentlib:jwp=transport=dt_socket,server=n,suspend=y,address=localhost:53730'  
ailsInExceptionMessages' '-cp' 'C:\Users\SOWJANYA\AppData\Local\Temp\vscode\ws_3581d\jdt_ws\jdt.ls-java-project\bin' 'SingletonPatternExample'  
Log: This is a log message from logger1.  
Log: This is a log message from logger2.  
1690716179  
1690716179  
Are both loggers the same instance? true  
PS C:\Users\SOWJANYA>
```

Exercise 2: Implementing the Factory Method Pattern

```
import java.util.Scanner;
```

```
public class FactoryMethodPattern {  
    public interface Document {  
        void open();  
        void close();  
    }  
    public static class WordDocument implements Document {  
        @Override  
        public void open() {  
            System.out.println("Opening Word Document.");  
        }  
        @Override  
        public void close() {  
            System.out.println("Closing Word Document.");  
        }  
    }  
    public static class PdfDocument implements Document {  
        @Override  
        public void open() {  
            System.out.println("Opening PDF Document.");  
        }  
        @Override
```

```

        public void close() {
            System.out.println("Closing PDF Document.");
        }
    }

    public static class ExcelDocument implements Document {
        @Override
        public void open() {
            System.out.println("Opening Excel Document.");
        }
        @Override
        public void close() {
            System.out.println("Closing Excel Document.");
        }
    }

    public abstract static class DocumentFactory {
        public abstract Document createDocument();
    }

    public static class WordDocumentFactory extends DocumentFactory {
        @Override
        public Document createDocument() {
            return new WordDocument();
        }
    }

    public static class PdfDocumentFactory extends DocumentFactory {
        @Override
        public Document createDocument() {
            return new PdfDocument();
        }
    }

    public static class ExcelDocumentFactory extends DocumentFactory {
        @Override

```

```

    public Document createDocument() {
        return new ExcelDocument();
    }
}

public static class FactoryMethodTest {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("Enter the type of document to create (Word, PDF, Excel):");
        String documentType = scanner.nextLine().trim().toLowerCase();
        DocumentFactory factory = null;
        switch (documentType) {
            case "word":
                factory = new WordDocumentFactory();
                break;
            case "pdf":
                factory = new PdfDocumentFactory();
                break;
            case "excel":
                factory = new ExcelDocumentFactory();
                break;
            default:
                System.out.println("Invalid document type.");
                System.exit(1);
        }
        Document document = factory.createDocument();
        document.open();
        document.close();
        scanner.close();
    }
}

```

```
}
```

Output:

```
PS C:\Users\SOWJANYA> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:51917'
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\SOWJANYA\AppData\Local\Temp\vscodesws_3581d\jdt_ws\jdt.ls-java-project\bin' 'FactoryMethodP
attern$FactoryMethodTest'
Enter the type of document to create (Word, PDF, Excel):
word
Opening Word Document.
Closing Word Document.
PS C:\Users\SOWJANYA>
```

```
PS C:\Users\SOWJANYA> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58195'
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\SOWJANYA\AppData\Local\Temp\vscodesws_3581d\jdt_ws\jdt.ls-java-project\bin' 'FactoryMethodP
attern$FactoryMethodTest'
Enter the type of document to create (Word, PDF, Excel):
excel
Opening Excel Document.
Closing Excel Document.
PS C:\Users\SOWJANYA>
```

```
PS C:\Users\SOWJANYA> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58222'
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\SOWJANYA\AppData\Local\Temp\vscodesws_3581d\jdt_ws\jdt.ls-java-project\bin' 'FactoryMethodP
attern$FactoryMethodTest'
Enter the type of document to create (Word, PDF, Excel):
pdf
Opening PDF Document.
Closing PDF Document.
PS C:\Users\SOWJANYA>
```